Nafis Karim

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Education

University of Illinois Chicago (UIC) Chicago, IL, USA
PhD in Computer Science *August 2025 – Present*
CGPA: 4.00/4.00

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering *May 2019 – June 2024*
CGPA: 3.81/4.00 (3.97 in final year, 3.93 in first two years)

Research Interest

Agentic AI Security, Behavioral & Browser Fingerprinting, Web Security, AI Agent Detection

Publications

- **AgentShield: Fortifying Web Applications Against AI Agents and Agentic Browsers**
  IEEE S&P, 2027
  [Accepted, with Shepherding]

  AgentShield presents a systematic empirical evaluation of modern AI browser agents (e.g., Perplexity Comet, ChatGPT Atlas, Claude for Chrome) against widely deployed anti-bot defenses, showing that many existing mechanisms fail to reliably distinguish autonomous agents from humans. We introduce a behavioral fingerprinting framework for detecting agentic clients, and identify Extinction-Induced Policy Drift, a novel failure mode in which agents can be driven to override their own safety constraints.

Teaching Experience

- **Teaching Assistant, CS 107 – Introduction to Computing and Programming** *Fall 2026*
  Department of Computer Science, University of Illinois Chicago
- **Teaching Assistant, Introduction to MATLAB** *Spring 2026*
  Department of Computer Science, University of Illinois Chicago
- **Teaching Assistant, CS 107 – Introduction to Computing and Programming** *Fall 2025*
  Department of Computer Science, University of Illinois Chicago
- **Lecturer, CSE Department** *July 2024 – August 2025*
  Brac University, Dhaka, Bangladesh
  Courses Taught: CSE321 (Operating Systems), CSE230 (Discrete Mathematics), CSE111 (OOP), CSE220 (Data Structures), CSE110 (Programming Language 1)

Research Experience

- **Enhancing Cyber Threat Intelligence (CTI) with Transformer-based LLMs: A Comparative Analysis with Few-Shot Learning** *05/23 – 12/23*
  Department of CSE, Bangladesh University of Engineering and Technology (BUET)
  Supervisor: Dr. Md. Shohrab Hossain
  Collaborators: Dr. Ying-Dar Lin, Dr. Ren-Hung Hwang (National Yang Ming Chiao Tung University)
  - Applied **Named Entity Recognition (NER)** to extract threat actors, malware, and other critical entities from unstructured Cyber Threat Intelligence (CTI) text.
  - Benchmarked transformer-based models (**BERT**, **RoBERTa**) against few-shot learning models (**GPT-3.5**, **GPT-4.0**) on the **DNRTI** and **APTNER** datasets.
  - Found transformer-based fine-tuning consistently outperforms few-shot prompting for specialized cybersecurity entity extraction.

- **Forecasting of COVID-19 Cases Using a Custom Deep Learning Architecture Incorporating Vaccination** *06/23 – 12/23*
  Department of CSE, Bangladesh University of Engineering and Technology (BUET)
  Award: Best Student Poster, NSysS 2023
  - Designed a fusion-type deep learning architecture combining channel-wise and global convolution with recurrent networks for time-series case forecasting.
  - Incorporated vaccination rates and hospital admission data, factors often excluded from prior forecasting models.
  - Outperformed state-of-the-art baselines on US and UK datasets, demonstrating vaccination's significant role in case trajectory.

Relevant Courseworks

Computer Security, Machine Learning, Artificial Intelligence, Operating Systems, Computer Networks, Software Engineering, Discrete Mathematics

Academic Projects

- **Neural Style Transfer using CNN** *11/23 – 02/24*
  Artistic style transfer inspired by Gatys et al., enabling style transfer from one image to another.
- **sub-C Compiler** *06/22 – 09/22*
  Developed a C compiler performing lexical, syntax, and semantic analysis, converting C code to x86 assembly.
- **Ray Tracing Project** *05/23 – 08/23*
  Generated realistic images of geometric shapes using ray tracing techniques.
- **Cryptography Algorithms, Security Vulnerabilities, and Attacks** *05/23 – 09/23*
  Explored malware, buffer overflow attacks, and browser exploitation using BeEF.
- **Nmap – Computer Security Project** *06/23 – 09/23*
  Explored various features of the network scanning tool, Nmap.
- **Non-invasive Mechanical Ventilator (Microcontroller Project)** *06/22 – 08/22*
  Designed and implemented a cost-effective non-invasive ventilator system.
- **Operating Systems Sessional** *12/22 – 02/23*
  Implemented various functionalities of the xv6 Operating System.

Technical Skills

- **Programming Languages:** Java, C++, Python, JavaScript
- **Security Tools:** Wireshark, NS2, CamFlow, Nmap
- **Frameworks:** Django, Node.js, React, PyTorch, TensorFlow
- **Markup Languages:** HTML, CSS, XML, LaTeX
- **Version Control:** Git

English Language Proficiency

TOEFL iBT: 111/120 (Speaking: 27, Writing: 27) • GRE: 160/170 (Verbal Reasoning)

Academic Recognition

- University Dean's List Scholarship (Levels 1, 2, 4)
- University Merit List Scholarship (Levels 1, 2, 4)
- Student Poster Champion, 8th International Conference on Networking, Systems and Security
- Student Poster Champion, 10th International Conference on Networking, Systems and Security